Reflection with Contribution

Name: Chris Roy

Contribution:

I answered Question 3 on the difference between Mean Squared Error (MSE) and accuracy, finalized the lab submission, and completed the reflection paper. I coordinated with team members to divide tasks, contributing primarily to the evaluation section, and ensuring all required files were included.

Reflection:

The lab highlighted the distinction between regression and classification. I clarified that MSE measures the average squared differences between predicted and true values, whereas accuracy reflects the proportion of correct classifications. I learned the importance of selecting appropriate evaluation metrics and the value of teamwork in task management and review processes, enhancing my understanding of model evaluation and communication in machine learning.

Camille: Reflection-This lab strengthened my understanding of how linear regression uses age and passenger class to predict ticket fares. Resolving a typing error and learning the importance of running code in the correct order sharpened my attention to detail and increased my confidence in using Python.

Contribution-I added a code cell print(lr_model.coef_) to display the model's coefficients and explained how age and passenger class relate to the predicted ticket fare, holding the other feature constant.

Min:

Reflection: In this lab, I learned how to make predictions on Titanic ticket fares with Linear Regression and our group used a Logistic Regression model to predict whether a passenger survived or not follow the same four-step process: create, fit, predict, and evaluate. While executing my code, I faced some issues of NameErrors due to commented lines and variable names not matching up, as well as seeing an inconsistency between the preset value of $48 in the template and the actual value from my model of $58.01. It surprised me to see that there was a difference of almost $58 since the average price of the ticket was only about $16, indicating that these two features (Age and Pclass) are not enough to make predictions about ticket fares on their own. Throughout fixing the issues I

learned how to consistently use this scikit-learn library in different projects since a notebook only stores variables when the relevant code in it runs.

Contribution:

For this lab, I prepared the lab notebook because the original file was unreadable and could not run. I then assigned roles to each member. I set up the notebook in Colab, shared it with everyone so that every member had access to work on it, and ran the data-preparation cells. For Task 1, I built and trained a Linear Regression model to predict Fare from Age and Pclass, then evaluated it with mean squared error. The result was an MSE of 3364.92, which means the model was off by about $58 on average. I also corrected the print statement in Task 1, which said the error was about $48, so that it matches the actual result of $58. Finally, I wrote the answer to Knowledge Check Q1 on the difference between regression and classification.

Name Rigoberto:

Reflection: In this lab, I learned the difference between regression and classification in machine learning. Regression is used to predict numerical values, while classification is used to predict categories. I practiced using Linear Regression to predict Titanic passenger fares and Logistic Regression to predict whether passengers survived. I also learned how to split data into training and testing sets and evaluate a model using Mean Squared Error and accuracy. One challenge was fixing the dataset URL and making sure the code ran correctly. This lab helped me understand how machine learning models use data to make predictions and how important it is to check their results.

Contribution: I worked on setting up the code, preparing the Titanic dataset, selecting the features, and training both machine learning models. I also worked on correcting errors in the code, making predictions, and evaluating the models. I answered the knowledge-check questions to explain what I learned about regression, classification, coefficients, and evaluation metrics. Through this lab, I improved my understanding of Python and the basic steps involved in building and testing machine learning models.